Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_17

1. Summary of Key Deviations

The analysis of Run_17 reveals a complex and severe fault scenario, combining a critical blockage with a significant pump over-performance.

- **Flow_Engine1 (Critical):** The most severe symptom is the total loss of flow to a single engine. The signal shows a **major negative drift** culminating in a complete **signal dropout (-99.7%)** in state s5.
- **Flow_Bypass (Major):** A very large positive **offset** is observed, with bypass flow reaching a massive **+58.4%** higher than normal. This indicates the vast majority of the system's flow is being diverted.
- **Flow_Pump (Moderate):** Unlike other blockage scenarios, the pump flow is not nominal but shows a very large positive **offset**, reading almost **+30%** higher than healthy. This indicates the pump is working extremely hard.
- **Other Signals (Nominal):** Crucially, `Flow_Engine2, 3, and 4` remain almost perfectly healthy, and there are no significant pressure anomalies or any activation of the `Flow_PRV` relief valve.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Injector Nozzle 1 & Main Fuel Pump**	**Dual Fault: Severe Blockage & Pump Over-performance**	This is the only hypothesis that fits all the evidence. A **severe blockage at Engine 1** explains the `Flow_Engine1` dropout and why `Flow_Bypass` is so high. Simultaneously, a separate fault is causing the **Main Fuel Pump to massively over-perform** (+30%). The combination of these two faults creates the observed scenario: the over-performing pump generates huge amounts of excess flow, and with the path to engine 1 blocked, almost all of this excess (plus the flow intended for engine 1) is forced through the bypass loop.

2	**Injector Nozzle 1**	**Severe Blockage / Clogging**	A blockage at Engine 1 alone would explain the `Flow_Engine1` dropout and the high `Flow_Bypass`. However, it does not explain why `Flow_Pump` is reading +30% higher than normal. In a simple blockage case, the pump flow should be closer to nominal.
3	**Main Fuel Pump**	**Severe Over-performance**	Pump over-performance alone would cause both `Flow_Pump` and `Flow_Bypass` to be very high. However, it does not explain why `Flow_Engine1` drops to zero while the other engines remain healthy. This proves there must also be a localized blockage.
4	**Flow_Engine1 Sensor & Flow_Pump Sensor**	**Dual Sensor Failure**	It's theoretically possible that the `Flow_Engine1` sensor failed (dropout) and the `Flow_Pump` sensor also failed (high offset) at the same time. However, the massive `Flow_Bypass` reading is physically consistent with the combination of these two events, making a dual physical fault more likely than a dual sensor fault.
5	**Bypass Valve**	**Stuck Open**	A stuck-open bypass valve would cause high bypass flow but would lead to a *drop* in pump flow and system pressure, which contradicts the observed +30% pump flow.

3. Most Probable Hypothesis

The most probable root cause is a **Dual Fault Scenario: (1) a Severe Blockage at Injector Nozzle 1 (or its feed line/FMU port) AND (2) a Significant Over-performance of the Main Fuel Pump.**

Reasoning: No single fault can explain the full set of symptoms. The data presents two distinct and overlapping problems. The dropout of `Flow_Engine1` while other engines are healthy is definitive proof of a localized blockage in that specific fuel line. However, the simultaneous, massive positive offset in `Flow_Pump` (+30%) and `Flow_Bypass` (+58%) cannot be explained by the blockage alone. This indicates that the pump itself is malfunctioning and delivering far too much fuel. It is the combination of these two independent failures that creates this unique and severe diagnostic signature.